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education

2026-2028 **HARVARD SCHOOL OF ENGINEERING AND APPLIED SCIENCES** **BOSTON, MA**
HARVARD BUSINESS SCHOOL
Candidate for Master in Science in Engineering Sciences and Master in Business Administration (joint MS/MBA), graduating 05/2028. Awarded the Dupré and Nunnelly Fellowship.

2018-2019 **PONTIFICIA UNIVERSIDAD CATOLICA DEL PERU** **LIMA, PERU**
Master of Science degree, *summa cum laude*, in Statistics, on a full-tuition scholarship. Thesis on Bayesian modeling of credit card utilization, presented at the Latin American Conference on Statistical Computing 2021.

2012-2016 **PONTIFICIA UNIVERSIDAD CATOLICA DEL PERU** **LIMA, PERU**
Bachelor of Science degree, *magna cum laude*, in Physics. Member of the Quantum Optics Research Group; selected for the Jorge André Swieca School on Nonlinear and Quantum Optics (Brazil, 2016). Awarded the PAIN-PUCP research grant. Founder and first President of the Physics Student Association.

research

Roa, L., Correa-Bahnsen, A., Suarez, G., Cortés-Tejada, F., et al. (2021). Super-App Behavioral Patterns in Credit Risk Models. *Expert Systems with Applications*, 169, 114486. 100+ citations.
A New Normal-Like Inflated Mean Regression for Bounded Responses (with Bayes and Valdivieso), working paper.

experience

2022-2026 **DUPPLA** **BOGOTA, COLOMBIA**
LatAm Series A fintech; first employee. Subject of a Harvard Business School case study on global entrepreneurship.
Founding Engineer

- ***Asset valuation:*** Built a home valuation ensemble (XGBoost, LightGBM; ~1M Bogotá homes); moved asset pricing and investor reporting from a 3-day manual process to real time, helping attract $1M+ in funding.
- ***Backend:*** Wrote the core platform in Python (FastAPI, PostgreSQL via SQLModel), scaling a monolith API into microservices with WebSockets for real-time transactions; ran it on GCP and AWS, saving $70K a year.
- ***Applied AI:*** Built an AI document validation engine, compressing legal review from 40 to 3 hours per deal and expanding financing capacity from 5 to 20 homes a month.

2022-2024 **PONTIFICIA UNIVERSIDAD CATOLICA DEL PERU** **LIMA, PERU**
Adjunct Professor (part-time)

- ***Teaching:*** Created a Computational Statistics course (R, Python) for 200+ students; faculty's top student rating.

2020-2022 **RAPPIBANK** **LIMA, PERU**
Founding team member at the digital bank built by Rappi, LatAm's largest super-app, and Interbank.
Machine Learning Engineer

- ***Credit risk modeling:*** Built the bank's first origination model, combining super-app behavioral data with credit bureau data on a 6TB platform; raised approvals for underserved customers 5% (published, see research).
- ***Fraud detection:*** Deployed XGBoost and PyTorch models in real time, halving inference latency to 100ms, reducing fraud losses 8% and lowering false positives 25%+.

2017-2020 **INTERBANK** **LIMA, PERU**
Peru's third-largest bank.
Data Scientist

- ***Credit loss provisioning:*** Built IFRS 9 expected credit loss models for ~3M credit cards and a S/ 60B loan book, using Markov transition matrices between delinquency buckets, stressed under macroeconomic scenarios.
- ***Loss simulation:*** Built a Monte Carlo credit loss simulator and logistic regression credit origination models.

projects

Real-time NBA arbitrage engine (Python): Odds scraper and automated execution bot; detects cross-book arbitrage from implied probabilities, sizes stakes to lock in ~2% typical edge, executes in under 30s (GitHub).

technical skills

Quantitative: probability, statistics, Bayesian modeling, Monte Carlo simulation, Markov models, machine learning
Tools: Python (NumPy, Pandas, SciPy, statsmodels, scikit-learn, PyTorch), R, SQL, PostgreSQL, GCP, AWS

certifications

Anthropic – Claude Certified Architect; Snowflake – SnowPro Core

personal

Fluent in English and Spanish. Rock climber (5.10c) and hiker (Inca Trail, Huayhuash, Torres del Paine).